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18 FEB 2018

A SMALL REMARK
ON LOOP
QUANTUM
GRAVITY.

05 MAY 2017

A SIMPLE AND
RIGOROUS
PROOF OF 26/10
DIMENSIONS IN
STRING THEORY

16 MAR 2017

GRAVITY,
ENTROPY, AND
LIFE.

20 FEB 2017

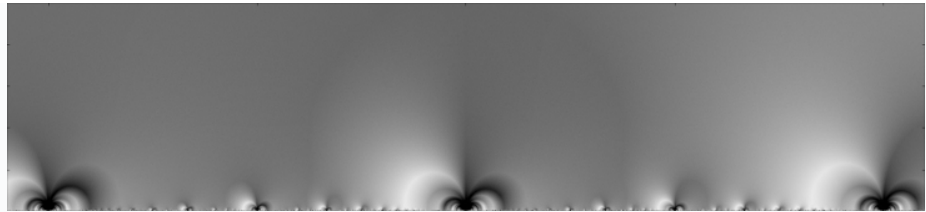
IS THE PLANCK
LENGTH THE
MINIMUM
POSSIBLE
LENGTH?

07 JAN 2017

TWIN PARADOX
FOR LITERAL
CHILDREN

07 SEP 2016

A GEOMETRIC
PROOF OF
HAWKING
RADIATION,



A SIMPLE AND RIGOROUS PROOF OF 26/10 DIMENSIONS IN STRING THEORY

05 MAY 2017

...does not exist.

At least, I couldn't manage. The requirement of a specific and large number of spacetime dimensions (26 for the simpler bosonic string theory, 10 for the more sophisticated superstrings) is one of the most misunderstood aspects of the theory and definitely the essential ingredient for the majority of negative gut feelings towards it. It seems like there should be more effort from string peeps to explain in the simplest way possible the origin of these bizarre numbers to non-experts.

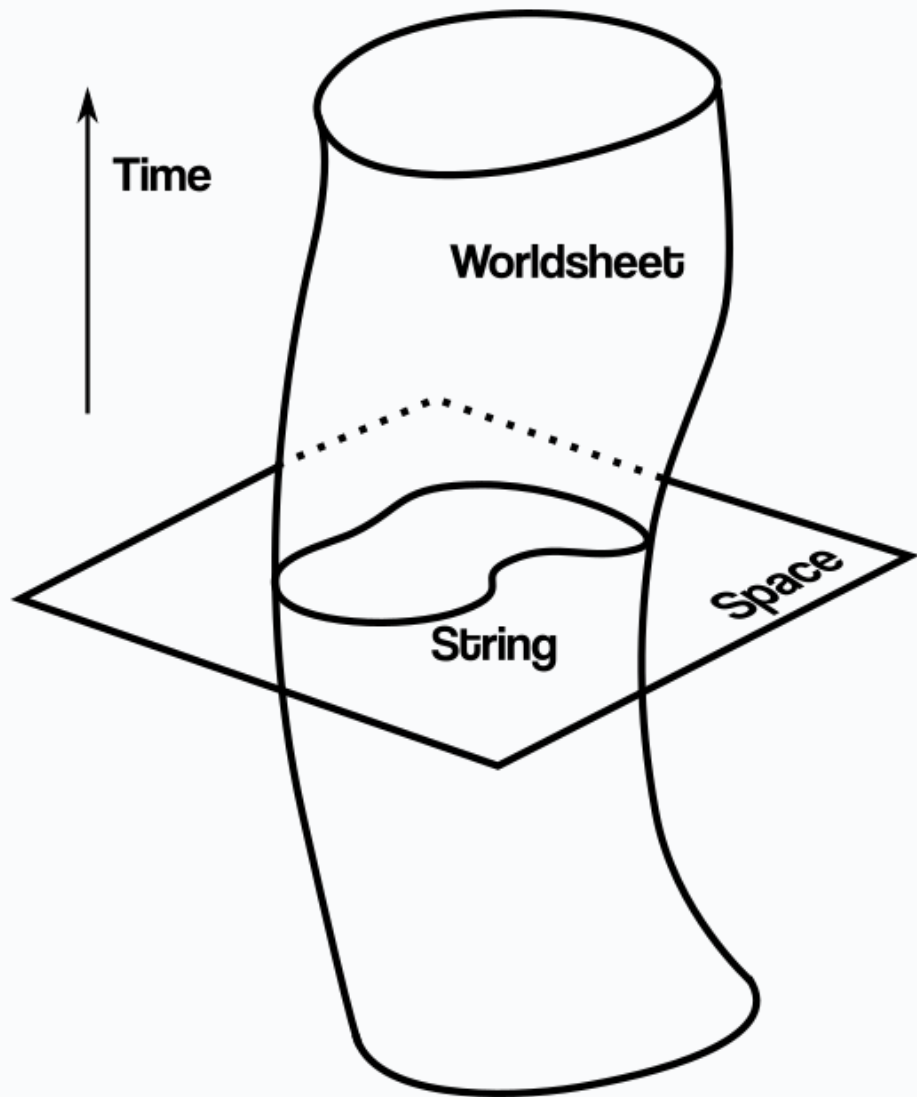
To be more specific, the real magical number is not the dimension D itself of spacetime, but $D - 2$, the number of dimensions **transverse** to the string, the ones it can oscillate *into* (minus one for the time dimension, and minus one for the dimension longitudinal to the string). In other words, the 1D string traces a 2D surface in spacetime, called a *worldsheet*. The magic number is the number of *remaining* directions available to the string, so $D - 2$.

THROUGH IMAGINARY TIME

30 MAY 2016

THE RIEMANN
ZETA FUNCTION,
THE PRIMON
GAS, AND
SUPERSYMMETRY.

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This picture has been badly drawn at least one thousand times, but I felt like badly drawing it yet one more time.

Thus the magic number is 24 (or 8 for the superstring). 24 is truly a mystical number. John Baez provides a [fantastic account](#) of why 24 is his favourite number, hinting at scattered mathematical gems in which it makes unexpected appearances. Some of them look like absolutely nothing else than pure numerology:

$$1^2 + 2^2 + 3^2 + \dots + 23^2 + 24^2 = 70^2$$

and [this only works](#) for 24, except 0 and 1 of course. (If you like **tough** math puzzles, try proving this. I don't). It is nothing else than incredible that this funny identity is related to unexpectedly complex and fascinating mathematics (monstrous moonshine), *and* string

theory (which acts as the “glue” that brings together monstrous moonshine appearance of $24/2$ in the crazy series

$$1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$$

In fact, that is why in bosonic string theory $D - 2 = 24$. In superstring theory, the same logic leads to $D - 2 = 10$, which is the critical dimension for superstring theory.

$$1 - 2 + 3 - 4 + \dots = \frac{1}{4}$$

which gives $D - 2 = 8$. You should at this point be convinced that I've lost all, these sound like random unrelated facts (and “facts”) and even though each can be easily explained to a layman, I'm not really explaining what the relationship is between the number of dimensions in string theory. I sound mad not because I say things that are crazy, but because these things are incoherent and disconnected. The problem is that the subject is too technical, either mathematically, or physically, or stringly, for me to explain in a way that is both accessible and accurate.

How to *convince* someone that $D - 2 = 24$ with the least effort possible? It's not possible to accept the crazy equation $1 + 2 + 3 + \dots = -1/12$; but that's clearly not the point. Even if it is possible to make sense of it, for example, with ζ -regularization, or analytic regularization + analytic renormalization (and there is already a lot of mathematics involved in this), this would *still* be unsatisfactory, since there is no non-shady reason why such a manipulation should have anything to do with the physics. Is it possible to derive the critical dimension *without* this crazy equation; actually, without ever encountering the need to regularize? Yes, sure. But exactly how **brief** and **elementary** can we be doing this?

I've found that proofs of $D = 26$ can be classified roughly as such:

1. Light-cone quantization

1. with crazy eqt. Requires a bit of knowledge about the polarization states of massive/massless vector bosons.
2. without crazy eqt. Requires essentially studying all of the string's Virasoro algebra.

2. Conformal field theory

1. without crazy eqt. Requires knowing conformal field theories and their anomalies.

3. Modular invariance

1. with crazy eqt. Requires fairly elementary QM and math.

2. without crazy eqt. Requires elementary QM but an f-ton of basic analysis.

Proof 1.1, I've done it [here](#) a while ago. It's pretty easy, but you need to trust me. The other proof with the crazy equation, 3.1, is in [Baez's slides](#).

The crazy-free proofs curiously form an unholy trinity. Each only requires one vertex of a triangle spanning between

1. String theory
2. Theoretical physics
3. Mathematics

for a certain (surely biased) interpretation of these terms. Proof 1.2 is the introductory string theory, since you're already learning the necessary string theory, it's mathematically not intimidating. Proof 2.1 is for a more advanced study of the physical interpretation of the critical number of dimensions as it is due to the cancellation of a conformal anomaly. Proof 3.2 I've never seen before - it's for people that know some basic complex analysis (residue theorem is all we need) and quantum mechanics, and requires next-to-no knowledge about string physics.

I've tried really hard to squeeze an **easy** proof that is "rigorous" (doesn't use fancy math) but I have failed. I have come to believe the three vertices of the triangle are three different translations in different languages of the same conceptual core, and that once two of the links in the chain are brought to a minimum complexity, the third

All I could manage is the following **realization of proof 3.2**. This will bring string physics to the background and will concentrate on the mathematical side as rigorous as possible on the mathematical side (while occasionally being slightly physics). It turns out quite longer than what you would call elementary, but that it builds the number 24 piece by piece (as 2, times 3, times 4). Consider the zeta function for releasing $1 + 2 + 3 + \dots = -1/12$ into the wild; if you care, here it is.

The proof

We have already talked about the worldsheet, the surface the string traces in spacetime. On this surface, we can set up a coordinate system (σ^1, σ^2) . Clearly, it is sensible to assume that the observables might be written in terms of these coordinates, they should

coordinate changes $(\sigma^1, \sigma^2) \rightarrow (\sigma^1 \iota, \sigma^2 \iota)$. After all, the coordinates are arbitrary. A simple example of (local) coordinate change is a dilation/rescaling:

$$(\sigma^1, \sigma^2) \rightarrow (\lambda \sigma^1, \lambda \sigma^2)$$

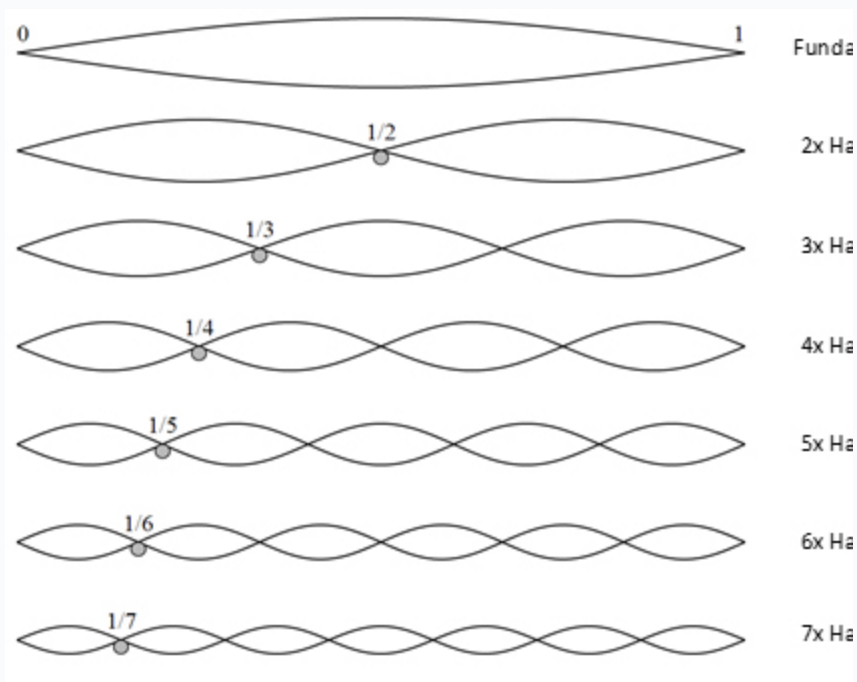
so we understand that our theory's observables must be **invariant under su** easy to do classically, but becomes highly non-trivial when the thing mechanical. This scale symmetry is part of the so-called worldsheet *cor* string theory.

Imagine a situation in which a string loops in time to trace a torus in spacetime. All possible tori shapes are all equally valid choices. We want to compute the probability for the process to happen with a specific torus shape, or better the *quantum amplitude*. The modulus is the probability. A basic point of quantum mechanics is that the probability amplitude is computed by summing $\exp(iE_i t)$ over all possible states i each with energy E_i after time passed t . Thus

$$Z = \sum_i e^{iE_i t}$$

is the total amplitude. Z should respect our desired symmetries.

An important point is that a composite system AB of two non-interacting su $Z_{AB} = Z_A Z_B$. Because of this nice property, we can first concentrate on a string oscillating in **one** single transverse dimension, and then take that to make it wiggle in $D - 2$ transverse dimensions.



Just like actual vibrating strings, string theory strings have harmonics which are ir fundamental.

So a string is almost exactly like an actual guitar strings and has infinite harmonics, or overtones. **Each one** of them is a harmonic oscillator, w/ version has energy levels

$$E_n = \omega \left(\frac{1}{2} + n \right)$$

Right? So the Z of one quantum harmonic oscillator is

$$Z_{QHO} = \sum_{n=0}^{\infty} e^{iE_n t} = e^{\frac{i}{2}\omega t} \sum_{n=0}^{\infty} e^{in\omega t} = \frac{e^{\frac{i}{2}\omega t}}{1 - e^{i\omega t}}$$

The geometric series I just summed appears to have ratio $|r| = 1$, which n not converge. Let's rip off the bandaid right now: the Z s, defined naively as converge in a Lorentzian (i.e. spacetime) theory. Often we physicists say because we sum a bunch of $\exp(ix)$ factors, and then handwave that they cancel, but this is a **cheap lie** - we just mean they don't converge. This is t theory but in all of quantum field theory, or standard quantum mechani quantum harmonic oscillator.

How can we move forward? The correct way to do it (and to word it) wi complex variable. By giving it an imaginary part, these Z s are made to conv you compute directly observable quantities from this, and then take t back recover sensible answers (which match experiment when possible). So, th prescription is our **definition** of what it means to have a quantum theory in s

tl;dr: don't worry, assume t is complex and in the upper half-plane $\text{Im } t > 0$.

To get back on track, we had that one string in one dimension had infini which clearly have frequencies which are integer multiples of the fu $\omega = 1, 2, 3, \dots$ and so the Z for our 1D string is the product

$$Z_{1,L} = \prod_{k=1}^{\infty} \frac{e^{\frac{i}{2}kt}}{1 - e^{ikt}} = e^{\frac{i}{2}(1+2+3+\dots)t} \left(\prod_k 1 - e^{ikt} \right)^{-}$$

...and we fell off the deep end. The madness of $1 + 2 + 3 + \dots$ caught up weak in spirit, we would succumb and substitute $1 + 2 + 3 + \dots \rightarrow -1/12$ the correct answer, skipping almost all of the math in this article. But we are here to make that 24 come out **without** using witchcraft. Therefore, let's

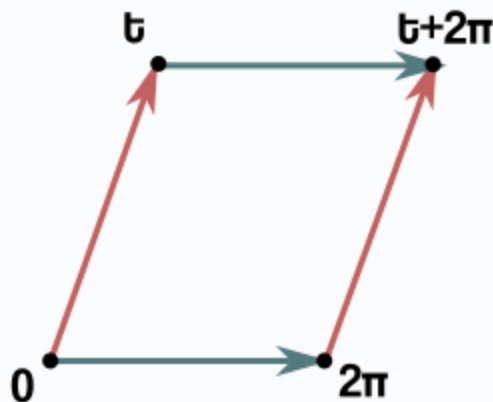
Why did that divergent sum appear in that exponent in the first place? Calculations back it comes from the zero-point energies of the QHOs, the zero-point energies are **arbitrary**, and that one was just a useful convention. There is an ambiguity in building a quantum HO from a classical HC ambiguity, since you need to convert commuting variables p, q into non-commuting $\hat{p}, \hat{q}$ and there is no preferential "quantization" of things like pq . Is the quantum $\hat{q}\hat{p}$? The difference is a constant, an arbitrariness in the zero-point energy. A conservative bet is that the ZPOs here actually sum to a finite, but unknown,

$$Z_{1,L}^{-1} = e^{-irt} \prod_k (1 - e^{ikt})$$

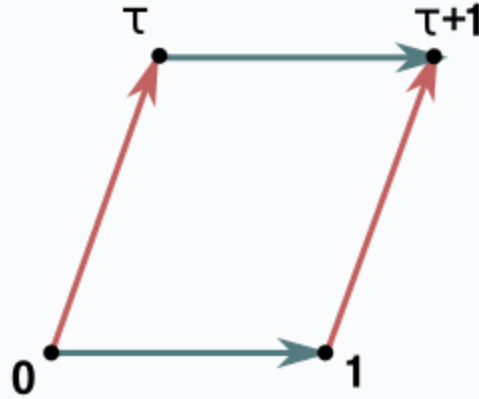
where r is an unknown real number. Note that the mad calculation would give $r = 1/24$. So, for my next two tricks:

- I am going to show, using a symmetry, that $1/r$ is the number of transverse dimensions.
- I am going to show, using another symmetry, that $r = 1/24$ anyway, even if $1 + 2 + 3 + \dots = -1/12$.

To start talking about these symmetries, I need to relate my variable t to the spatial coordinate x . The relationship is that the torus is built by gluing same-colour edges in this



where t is represented as a point in the complex plane. (The 2π side is t convention that made it so that the frequencies $\omega = 1, 2, 3, \dots$ were integer notation, with $\tau = t/2\pi$ (and a simple rescaling, which I remind is a symme



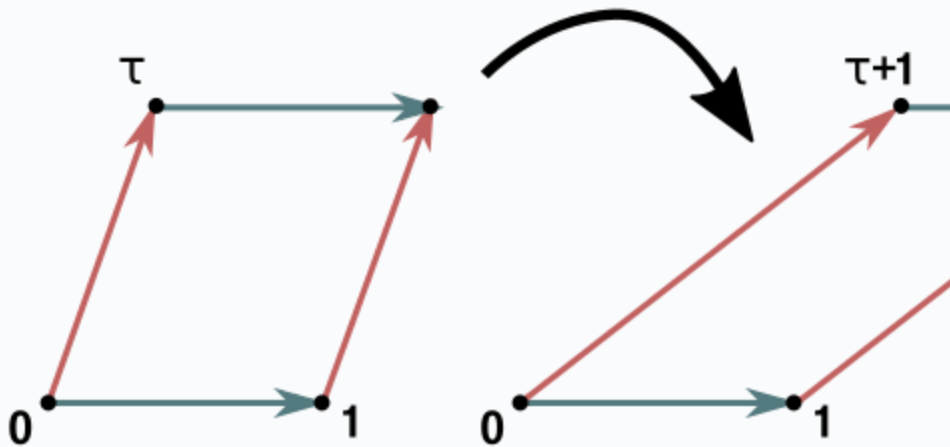
Note that when τ moves to the real axis the torus degenerates. That makes string really loop back unto itself in real time? That'd be time travel. It imaginary time or at least in the upper half-plane.

Defining $q := e^{i2\pi\tau}$, our $Z_{1,L}$ becomes

$$Z_{1,L}^{-1}(\tau) = q^r \prod_k (1 - q^k)$$

First trick

First of all, if we shift $\tau \rightarrow \tau + 1$, we get different parallelograms:



but they make the same torus after gluing (check it!). If the shape does not the physical quantities shouldn't either.

However, under this transformation our amplitude does transform:

$$Z_{1,L}^{-1} \rightarrow e^{2\pi i r} Z_{1,L}^{-1}$$

That makes no sense... until we remember that this is only for **one** di-
transverse dimensions, the total amplitude is $Z_{1,L}^{D-2}$, which will remain invariant

$$r(D-2) = 1$$

this equation defines the *critical dimension* of the string theory. If we will find $1/r$, then the theory will make sense only in $D = 1/r + 2$ spacetime dimensions.

Now, proving that $r = 1/24$ will not be as easy. Let's try.

Second trick

The other symmetry we will consider is to send $\tau \rightarrow -1/\tau$. The resulting parallelogram is similar to the original one but they are rescaled versions of each other. You can check that the side matches with the new $-1/\tau$ side and the original τ side matches with the new 1 side. In other words, rescalings of the worldsheet should be a symmetry.

Thus the total amplitude must be invariant under this; however we should also be invariant under the first transformation. Huh? Wasn't $Z_{1,L}^{D-2}$ the total amplitude? No, I lied to you.

harsh truth. The truth (part of it) is that **oscillation on a string always comes in** many possible characterizations of this dichotomy: left-movers and right-movers, holomorphic and antiholomorphic, sine and cosine...

What matters is that to our $Z_{1,L}$ (the L is now understood to stand for “left” paired $Z_{1,R}$ for the sister oscillations. Thankfully, I’ll just handwave that $Z_{1,L}$ overall amplitude is the product $|Z_{1,L}^{D-2}|^2$.

However, that is still not all: while we have accounted for oscillations of the “reference” position, we also need to account for the *overall* movement of the string in space. We want the string to be back to its original point after time τ if we wrap around the torus; but we cannot just *want* it, we need to include the probability amplitude

What is the probability amplitude for a quantum particle in 1D to stay in the same position after a certain amount of time? A hint is that for **purely imaginary times**, the Schrödinger equation is equivalent to the [heat equation](#). And an infinitely concentrated speck of heat evolves in 1D by [spreading into a gaussian whose peak decreases as](#) $(\text{time})^{-1/2}$. The probability amplitude for the centre of mass is something like this

$$(\text{Im } \tau)^{-1/2}$$

If so, putting everything back together the total amplitude should be something like

$$Z \sim (\text{Im } \tau)^{-\frac{D-2}{2}} |Z_{1,L}|^{2(D-2)}$$

Ok, now under $\tau \rightarrow -1/\tau$,

$$\text{Im } \tau \rightarrow \frac{1}{|\tau|^2} \text{Im } \tau$$

therefore if $Z_{1,L}$ were to transform in some way similar to this:

$$Z_{1,L} \rightarrow \tau^{-1/2} Z_{1,L}$$

our total amplitude would be invariant, and our theory would make sense. The only way that this can only happen is if $r = 1/24$.

So, sorry

Let’s do some preliminary rewriting.

$$Z_{1,L}^{-1}(\tau) = q^r P(\tau), \quad P(\tau) := \prod_{\ell=1}^{\infty} (1 - q^{\ell})$$

Then, a more comfortable presentation of the product $P(\tau)$:

$$-\log P(\tau) = -\sum_{\ell=1}^{\infty} \log(1 - q^{\ell}) = \sum_{l,k=1}^{\infty} \frac{1}{k} q^{k\ell} = \sum_{k=1}^{\infty} \frac{1}{k} \frac{q^k}{1 - q^k} = \sum_{k=1}^{\infty} \frac{q^k}{k(1 - q^k)}$$

I've used the Taylor expansion for $-\log(1 - x)$ and the geometric series. I'm convinced, you can check more carefully these steps (including the sum $\sum_{k=1}^{\infty} \frac{q^k}{k(1 - q^k)}$ for $\text{Im } \tau > 0$).

Now:

The following psychedelic argument is due to Siegel (yeah, that Siegel). We reason from this function of a complex variable w , with τ as a parameter:

$$f(w) = \cot w \cot \frac{w}{\tau}$$

and then, introducing another real parameter ν (our finishing move with infinity), we construct the combination

$$g(w) = \frac{f(\nu w)}{w}$$

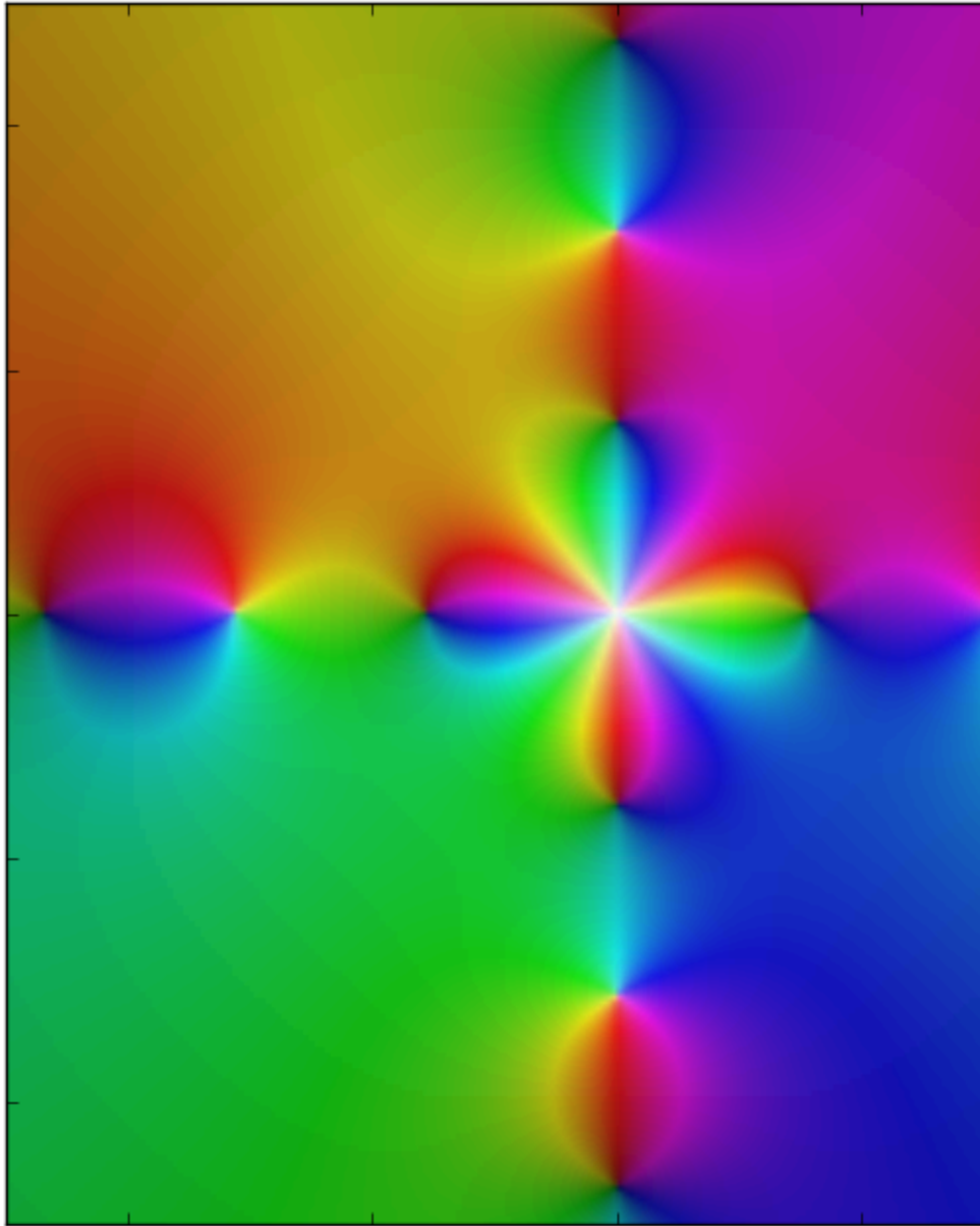
Let's count the poles of g . A quick inspection shows there is a set of simple poles at $w = \pm \frac{\pi k \tau}{\nu}$, for $k = 1, 2, 3, \dots$; plus a triple pole at $w = 0$. The residues are computed as follow:

$$\text{Res}_{\pm \frac{\pi k}{\nu}} g = \frac{1}{\pi k} \cot\left(\frac{\pi k}{\tau}\right)$$

$$\text{Res}_{\pm \frac{\pi k \tau}{\nu}} g = \frac{1}{\pi k} \cot(\pi k \tau)$$

$$\text{Res}_0 g = -\frac{1}{3}(\tau + \tau^{-1})$$

If the residue at the triple point doesn't seem obvious, recall that the Taylor expansion of the cotangent at zero starts $\cot s \sim \frac{1}{s} - \frac{s}{3}$.



Colour wheel plot of $(g(w))$, for $(\tau = i)$, $(\nu = 1)$. The white specks are the poles, many times the colours repeat circling them.

Now, it should be clear the intention is to exploit g for an application of the residue theorem. Consider the path γ running ccw around the parallelogram with vertices $1, \tau$

The residue theorem says

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(\nu w)}{w} d\omega = \sum_{p \in \text{poles}} \text{Res}_p g$$

The sum is actually only over those poles that are inside the parallelogram, while that's not something we should worry about.

Now, I would like to use this equation to prove the identity we actually need, however, for τ on the imaginary axis, because it's simpler, but it will also work for $\text{Im } \tau > 0$. Since both sides of the equation I'll derive are holomorphic in the τ plane, them agreeing on a line is enough to prove they are always equal. Long τ is purely imaginary for now, but at the end we can just drop this assumption.

Let's do the integral on the parallelogram, which is now a rhombus. $g(w)$ seems to be an easy function to guess an antiderivative for. So let's take the derivative of $\log w$. It's hard to see that $f(\nu w)$ converges to the constant $(1, -1, 1, -1)$ resp segments of γ (if you don't see it, write $\cot s$ using complex exponentials). The integral is

$$\left(\int_1^\tau - \int_\tau^{-1} + \int_{-1}^{-\tau} - \int_{-\tau}^1 \right) \frac{dw}{w}$$

Easy! The integral of $1/w$ is $\log w$, so let me just plug that in and... aaagh! I choose a branch for the log, and make sure the path does not jump over the branch cut. I use symmetry to rewrite

$$2 \left(\int_1^\tau + \int_1^{-\tau} \right) \frac{dw}{w}$$

and now that the path is all on the right, we can use the negative real axis as well. I use the principal branch of the logarithm. Keeping track of everything at the end gives

$$4 \log\left(\frac{\tau}{i}\right)$$

Nice. Now for the residues. If $\nu \rightarrow \infty$, all the poles shrink and move closer to the origin. In the limit they are all inside the rhombus and the sum is over all poles. Thus the sum is

$$-\frac{1}{3}(\tau + \tau^{-1}) + 2 \sum_{k=1}^{\infty} \left(\frac{1}{\pi k} \cot\left(\frac{\pi k}{\tau}\right) + \frac{1}{\pi k} \cot(\pi k \tau) \right)$$

Note

$$\cot s = \frac{e^{is} + e^{-is}}{e^{is} - e^{-is}} = \frac{1 + e^{-2is}}{1 - e^{-2is}} = -1 + \frac{2}{1 - e^{-2is}}$$

plug that in, and you'll recognize that the sum of the residues becomes

$$\begin{aligned}
& -\frac{1}{3}(\tau + \tau^{-1}) + \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{1}{1 - e^{-\frac{2\pi i k}{\tau}}} - \frac{1}{1 - e^{-2\pi i k \tau}} \right) \\
& = -\frac{1}{3}(\tau + \tau^{-1}) + \frac{2}{\pi} (\log P(-1/\tau) - \log P(\tau))
\end{aligned}$$

Finally we're starting to get back to Earth. This is starting to look like a theorem product. Now that we have LHS and RHS, let's use the residue theorem.

$$\frac{1}{2} \log\left(\frac{\tau}{i}\right) = -\frac{\pi i}{12}(\tau + \tau^{-1}) - (\log P(\tau) - \log(P(-1/$$

There's our magic number! Half of it at least. This still looks like gibberish, but if we exponentiate for good measure:

$$e^{\frac{2\pi i}{24} \tau} P(-1/\tau) = \sqrt{\tau/i} e^{\frac{2\pi i \tau}{24}} P(\tau)$$

This seems not much of a fact about $P(\tau)$, but about the combination

$$\eta(\tau) := e^{\frac{2\pi i \tau}{24}} P(\tau) = q^{\frac{1}{24}} \prod_{k=1}^{\infty} (1 - q^k)$$

which we just proved transforms “nicely” under a $\tau \rightarrow -1/\tau$ transformation

$$\eta(-1/\tau) = \sqrt{\frac{\tau}{i}} \eta(\tau)$$

And that's it! That's what we were looking for. For $Z_{1,L}^{-1}(\tau)$ to transform how $\eta(\tau)$ itself, and so $r = 1/24$, and so, finally

$$D = 26$$

Conclusions

Because of the philosophy behind this whole post, to only use mathematics as possible and to have it be self-contained, we sped by what is actually incredibly much more elegant) math. $\eta(\tau)$ is, of course, the [Dedekind eta function](#) and its transformation properties under the modular group; moreover, the modular discriminant $\Delta(\tau) := (2\pi)^{12} \eta^{24}$ is a modular form of weight 12. So, if you're connected to this area of mathematics; in fact I hope that for those that are

this served as a sneak peek of what strings even have to do with modular forms. I think I would do justice to a subject I actually barely know the basics of, so I'll

An interesting question that pops up sometimes: if it's so $1 + 2 + 3 + \dots = -1/12$, or ζ -regularization / heat kernel / Abel summation, want to call it, give the same correct result as more rigorous paths, with one

I have **no** idea.

Instinctively I would babble something about “physically reasonable” or “arithmetic” but this is just pure madness. It's definitely not a coincidence because the same is true for superstrings too. In fact I'm not even sure what the exact relationship between these two classes of proofs are, they look like completely different reasonings. In light of this, on a first reading you don't even realize how the scale/conformal invariance is hidden. And if you happen to know some conformal field theory, go take a look at how it builds the 24. They don't sound *exactly* like translations of the same thing, even though they are to be.

It's fascinating how much information about the content of the theory the $D = 10$ carry, and how much you already learn about strings just by trying to understand essentially the **most basic** fact about the theory. It is a limpid example of a theory *everything fits*, nothing is just thrown there, everything is essential.

Addendum

There is perhaps a faster proof if you assume Euler's pentagonal number theorem:

$$P(\tau) = \prod_{k=1}^{\infty} (1 - q^k) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}$$

If you complete the square in the exponent you can rewrite

$$P(\tau) = q^{-1/24} \sum_{k=-\infty}^{\infty} (-1)^k q^{\frac{3}{2}(k-\frac{1}{6})^2}$$

thus

$$\eta(\tau) = q^{1/24} P(\tau) = \sum_{k=-\infty}^{\infty} (-1)^k q^{\frac{3}{2}(k-\frac{1}{6})^2}$$

With η in this form, it is *doable* to prove $\eta(-1/\tau) = \sqrt{\tau/i}\eta(\tau)$ using Poisson

Still, invoking Euler's theorem seems definitely like cheating, so I didn't pursue

Refs

Baez, J. (2008). [My favourite number: 24](#). Talk at the University of Glasgow.

Siegel, C. (1954). A simple proof of $\eta(-1/\tau) = \eta(\tau)\sqrt{\tau/i}$. *Mathematics of the*
doi:10.1112/S00255793000000462

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